

WASP-10b

R-0903

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_240702 · created 2026-07-29T03:59:19+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460493.8418 +/- 0.0064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.03 +/- 0.062
Transit depth (Rp/Rs)^2	0.089 +/- 0.37 [%]
Orbital Inclination (inc)	83.86 +/- 1.85
Airmass coefficient 1 (a1)	0.69 +/- 0.4
Airmass coefficient 2 (a2)	0.2 +/- 0.15
Scatter in the residuals of the lightcurve fit is	56.9 %
Best Comparison Star	#1 - [154, 360]
Optimal Aperture	3.03
Optimal Annulus	3.29
Transit Duration (day)	0.016 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.03 ± 0.062 vs 0.1592 (deviation 2.08σ)
Duration fit vs published	0.016 d vs 0.0946 d (deviation 3.02σ)
Mid-time O–C	16.6 min
Depth SNR	0.5
Residual scatter	56.9 %

Light curve

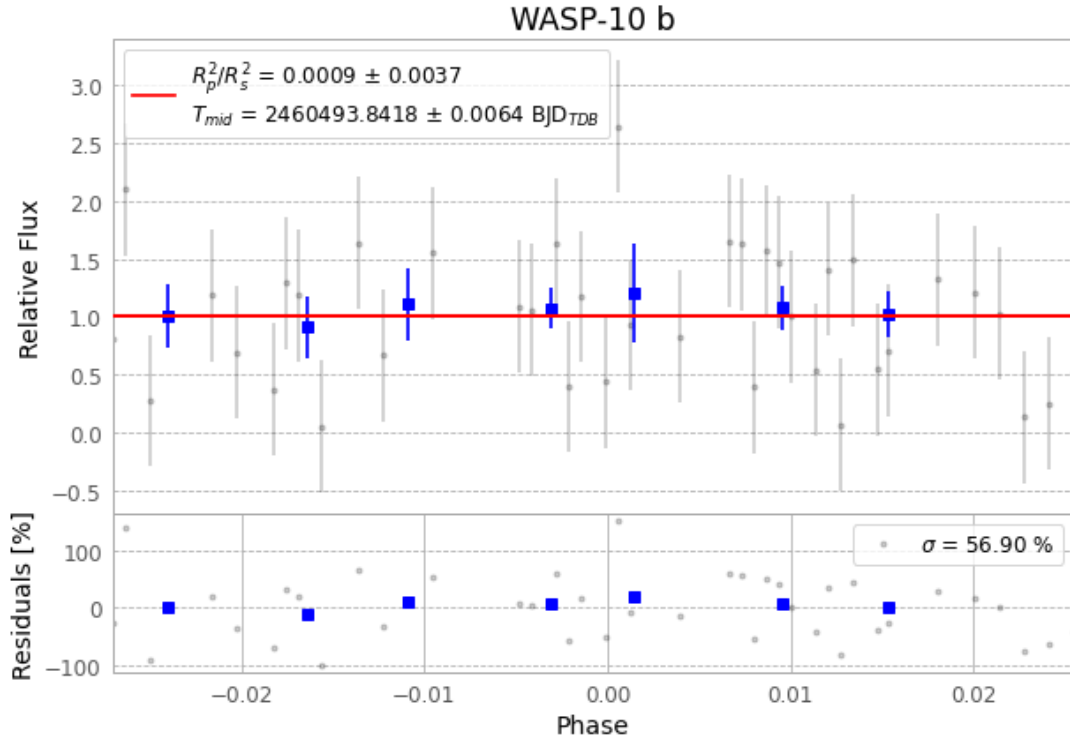


Figure 1 — FinalLightCurve_WASP-10 b_2024-07-01.png (SHA-256 021c84ba0b1c6717...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [237, 220], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6bf60cb0c41d5f60...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

84 FITS frames (55 MB), WASP-10240702055415.FITS → WASP-10240702100315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-240702.log (90fd44bc1031...), FinalLightCurve_WASP-10 b_2024-07-01.png (021c84ba0b1c...), FinalLightCurve_WASP-10 b_2024-07-01.csv (c63759f901b9...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 03:59Z.